Computer Networks LAB (CS3P004)
Spring Semester 2024-25
Lab Sheet 2

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Ans – 1) A packet sniffer targets and captures packets of data transmitting over a network. It is similar to wire tapping over a telephone network. It can be used to identify network problems, but on the other hand, can also be used by hackers to collect information.

Wireshark – It is a packet analyzer (also known as packet sniffer) program. It can be used to troubleshoot network issues, ensure proper layout of protocols and identify possible security threats.

Ans – 2)

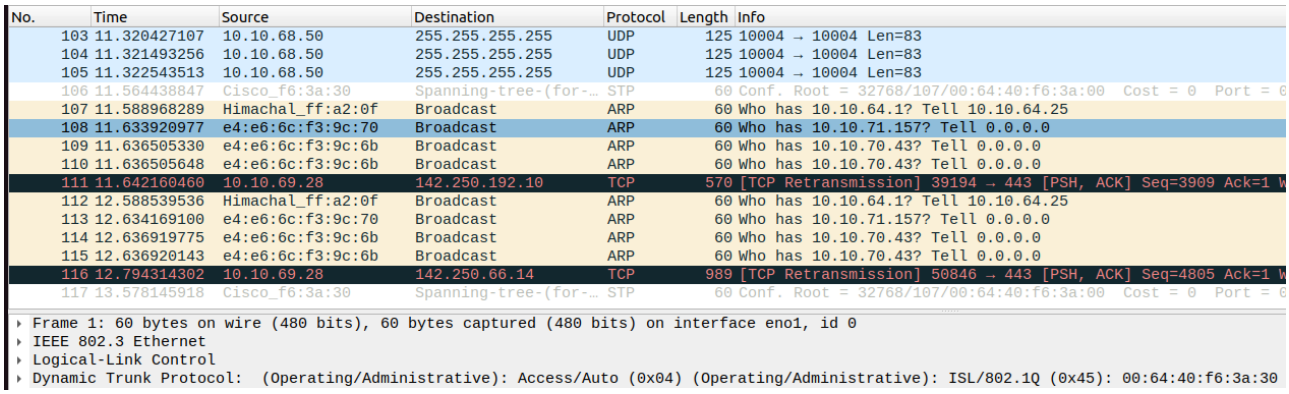
a) Interface of "Capture" section in **Wireshark** looks somewhat similar to the **traceroute** command where for each packet we can see its latency and hops. Also, it tells us the IP of network similar to the **nslookup** command.

b) **Livecapture -** It allows us to capture packets in real time and and provide information about those packets. It has "filter" command where we can filter out a certain section if we want to specifically analyze that part.
It captures the network packets from different network media types such as Ethernet, Wireless LAN, Bluetooth, USB Tethering, Wi-Fi and others.

c) Sections of WireShark include -
   i. Find a packet
   ii. Time Shift
   iii. Go to previous, next or a certain packet.
   iv. Start a **Live Capture**
   v. Analyze packets
   vi. Statistics
   vii. Telephony
   viii. Wireless (includes Bluetooth)
   ix. WLAN Traffic
   x. Firewall Rules and Credentials

Ans – 3)

a)



No.	Time	Source	Destination	Protocol	Length	Info
103	11.320427107	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
104	11.321493256	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
105	11.322543513	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
106	11.564438847	Cisco_f6:3a:30	Spanning-tree-(for-…	STP	60	Conf. Root = 32768/107/00:64:40:f6:3a:00 Cost = 0 Port = 6
107	11.588968289	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
108	11.633920977	e4:e6:6c:f3:9c:70	Broadcast	ARP	60	Who has 10.10.71.157? Tell 0.0.0.0
109	11.636505330	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
110	11.636505648	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
111	11.642160460	10.10.69.28	142.250.192.10	TCP	570	[TCP Retransmission] 39194 → 443 [PSH, ACK] Seq=3909 Ack=1 W
112	12.588539536	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
113	12.634169100	e4:e6:6c:f3:9c:70	Broadcast	ARP	60	Who has 10.10.71.157? Tell 0.0.0.0
114	12.636919775	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
115	12.636920143	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
116	12.794314302	10.10.69.28	142.250.66.14	TCP	989	[TCP Retransmission] 50846 → 443 [PSH, ACK] Seq=4805 Ack=1 W
117	13.578145918	Cisco_f6:3a:30	Spanning-tree-(for-…	STP	60	Conf. Root = 32768/107/00:64:40:f6:3a:00 Cost = 0 Port = 6

```
▸ Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eno1, id 0
▸ IEEE 802.3 Ethernet
▸ Logical-Link Control
▸ Dynamic Trunk Protocol:  (Operating/Administrative): Access/Auto (0x04) (Operating/Administrative): ISL/802.1Q (0x45): 00:64:40:f6:3a:30
```

b) The familiar keywords here are:

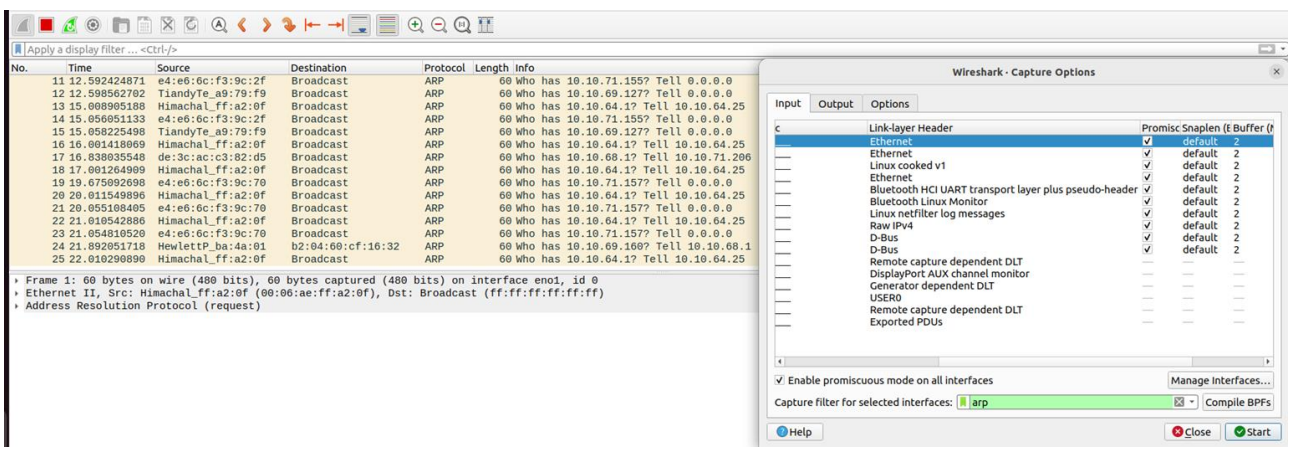
- ➢ Time (**same as traceroute)**
- ➢ Source IP
- ➢ Destination IP
- ➢ Hops
- ➢ Packet No.

c) **Display filter –** Captures all the packets irrespective of whether they are included in the filter or not and then analyzes the only a certain section of the filter that are to be filtered.

Capture Filter – Captures only the set of packets included on the filter and disregards all the other packets and they are not saved.

Hence, even though both of them might output the same analysis, Capture Filter saves on space due to it not saving unused data.

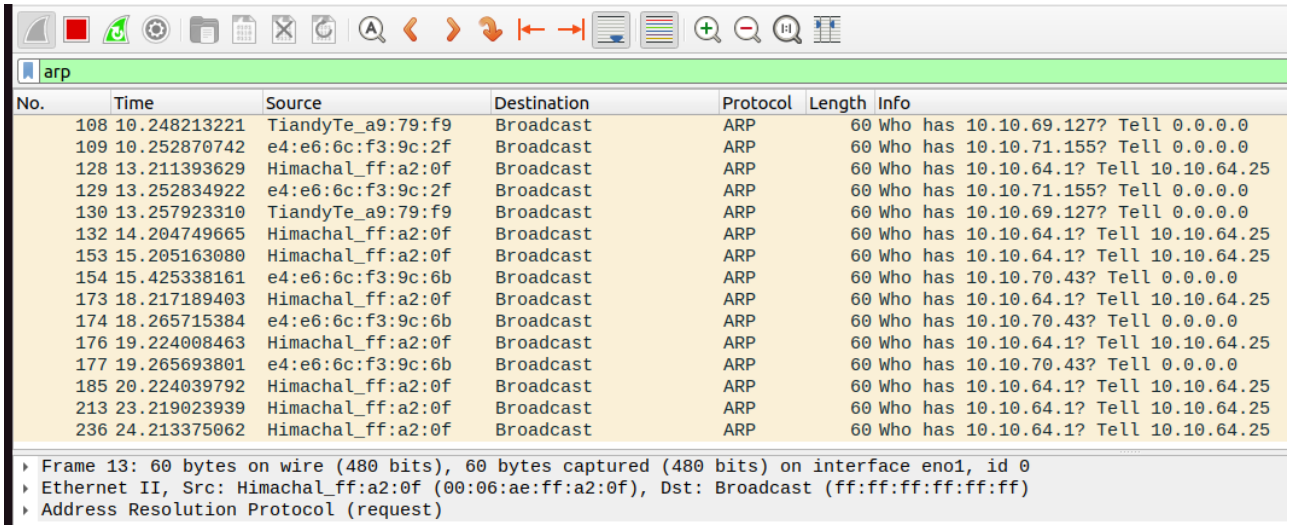
i. **Capture Filter** results -



No.	Time	Source	Destination	Protocol	Length	Info
11	12.592424871	e4:e6:6c:f3:9c:2f	Broadcast	ARP	60	Who has 10.10.71.155? Tell 0.0.0.0
12	12.598562702	TiandyTe_a9:79:f9	Broadcast	ARP	60	Who has 10.10.69.127? Tell 0.0.0.0
13	15.008905188	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
14	15.056051133	e4:e6:6c:f3:9c:2f	Broadcast	ARP	60	Who has 10.10.71.155? Tell 0.0.0.0
15	15.058225498	TiandyTe_a9:79:f9	Broadcast	ARP	60	Who has 10.10.69.127? Tell 0.0.0.0
16	16.001418069	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
17	16.838035548	de:3c:ac:c3:82:d5	Broadcast	ARP	60	Who has 10.10.68.1? Tell 10.10.71.206
18	17.001264909	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
19	19.675092698	e4:e6:6c:f3:9c:70	Broadcast	ARP	60	Who has 10.10.71.157? Tell 0.0.0.0
20	20.011549896	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
21	20.055108405	e4:e6:6c:f3:9c:70	Broadcast	ARP	60	Who has 10.10.71.157? Tell 0.0.0.0
22	21.010542886	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
23	21.054810520	e4:e6:6c:f3:9c:70	Broadcast	ARP	60	Who has 10.10.71.157? Tell 0.0.0.0
24	21.892051718	HewlettP_ba:4a:01	b2:04:60:cf:16:32	ARP	60	Who has 10.10.69.160? Tell 10.10.68.1
25	22.010290890	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25

```
▸ Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eno1, id 0
▸ Ethernet II, Src: Himachal_ff:a2:0f (00:06:ae:ff:a2:0f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (request)
```

Display Filter results -



arp

No.	Time	Source	Destination	Protocol	Length	Info
108	10.248213221	TiandyTe_a9:79:f9	Broadcast	ARP	60	Who has 10.10.69.127? Tell 0.0.0.0
109	10.252870742	e4:e6:6c:f3:9c:2f	Broadcast	ARP	60	Who has 10.10.71.155? Tell 0.0.0.0
128	13.211393629	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
129	13.252834922	e4:e6:6c:f3:9c:2f	Broadcast	ARP	60	Who has 10.10.71.155? Tell 0.0.0.0
130	13.257923310	TiandyTe_a9:79:f9	Broadcast	ARP	60	Who has 10.10.69.127? Tell 0.0.0.0
132	14.204749665	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
153	15.205163080	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
154	15.425338161	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
173	18.217189403	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
174	18.265715384	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
176	19.224008463	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
177	19.265693801	e4:e6:6c:f3:9c:6b	Broadcast	ARP	60	Who has 10.10.70.43? Tell 0.0.0.0
185	20.224039792	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
213	23.219023939	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
236	24.213375062	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25

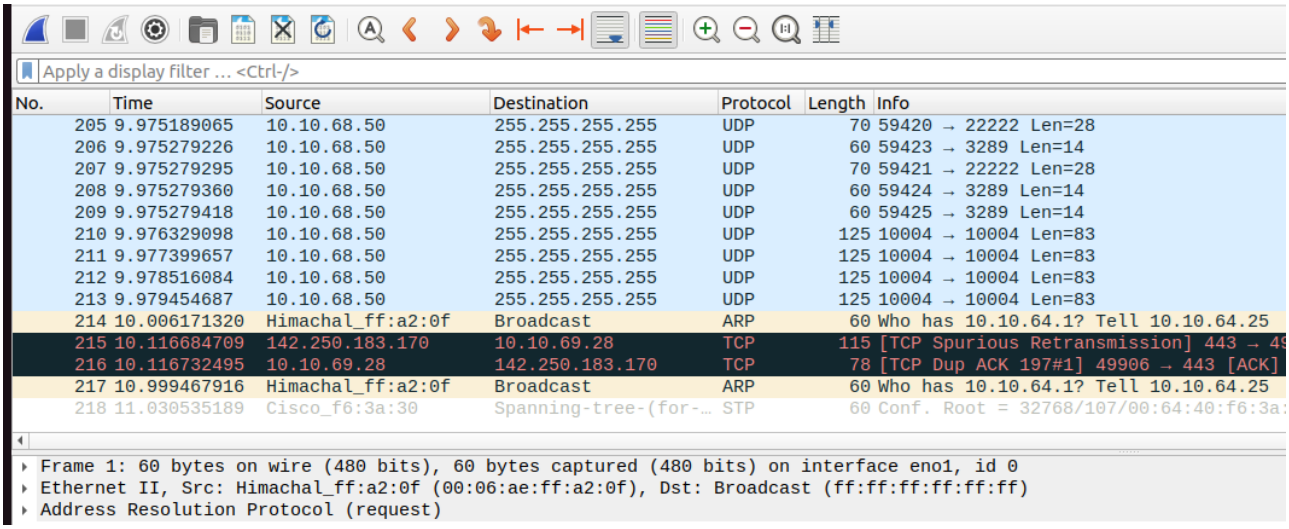
```
▸ Frame 13: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eno1, id 0
▸ Ethernet II, Src: Himachal_ff:a2:0f (00:06:ae:ff:a2:0f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (request)
```

Major difference – Observe that packet numbers after putting the "arp" as Capture Filter is in order, whereas in Display Filter after putting "arp" the order is random, this is because display filter captures other packets and the Packet no. Shown here includes those of other non-arp captures packets.

Ans – 4)

a) Steps:

i. Stopped Live Capture

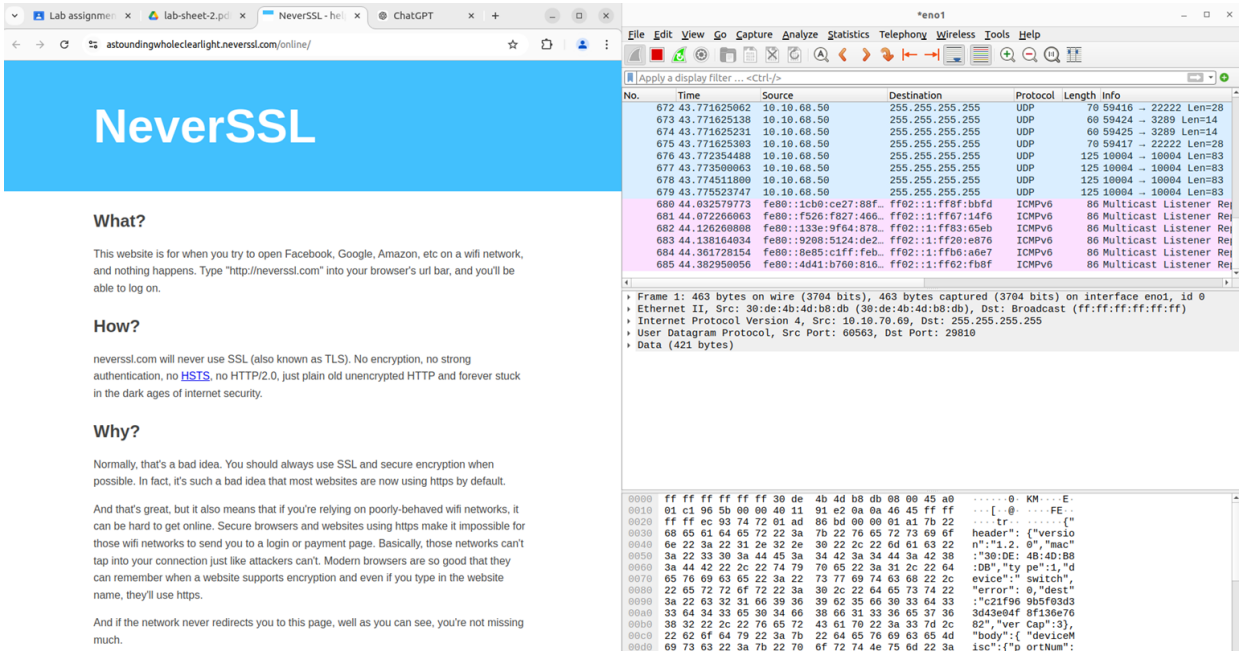


Apply a display filter … <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
205	9.975189065	10.10.68.50	255.255.255.255	UDP	70	59420 → 22222 Len=28
206	9.975279226	10.10.68.50	255.255.255.255	UDP	60	59423 → 3289 Len=14
207	9.975279295	10.10.68.50	255.255.255.255	UDP	70	59421 → 22222 Len=28
208	9.975279360	10.10.68.50	255.255.255.255	UDP	60	59424 → 3289 Len=14
209	9.975279418	10.10.68.50	255.255.255.255	UDP	60	59425 → 3289 Len=14
210	9.976329098	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
211	9.977399657	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
212	9.978516084	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
213	9.979454687	10.10.68.50	255.255.255.255	UDP	125	10004 → 10004 Len=83
214	10.006171320	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
215	10.116684709	142.250.183.170	10.10.69.28	TCP	115	[TCP Spurious Retransmission] 443 → 49
216	10.116732495	10.10.69.28	142.250.183.170	TCP	78	[TCP Dup ACK 197#1] 49906 → 443 [ACK]
217	10.999467916	Himachal_ff:a2:0f	Broadcast	ARP	60	Who has 10.10.64.1? Tell 10.10.64.25
218	11.030535189	Cisco_f6:3a:30	Spanning-tree-(for-…	STP	60	Conf. Root = 32768/107/00:64:40:f6:3a:

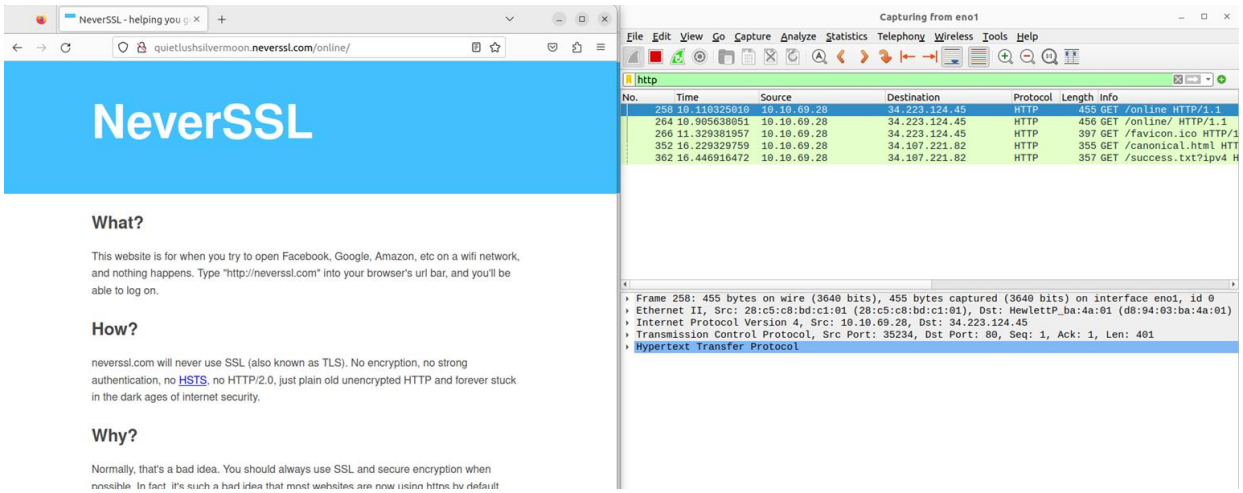
```
▸ Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eno1, id 0
▸ Ethernet II, Src: Himachal_ff:a2:0f (00:06:ae:ff:a2:0f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▸ Address Resolution Protocol (request)
```

ii. Start Capture + Opened Browser http://neverssl.com



iii. 'http' in display filter

Here "Mozilla Firefox" is used to get more "http" packets.



b) “http” Request packets:

http

No.	Time	Source	Destination	Protocol	Length	Info
124	5.608595103	10.10.69.28	34.223.124.45	HTTP	398	GET / HTTP/1.1
149	5.711041772	10.10.69.28	34.223.124.45	HTTP	465	GET /online HTTP/1.1
177	6.528463275	34.223.124.45	10.10.69.28	HTTP	319	HTTP/1.1 301 Moved Permanently (text/html)
179	6.531418794	10.10.69.28	34.223.124.45	HTTP	466	GET /online/ HTTP/1.1
183	6.889958364	34.223.124.45	10.10.69.28	HTTP	1227	HTTP/1.1 200 OK (text/html)
185	6.926564887	10.10.69.28	34.223.124.45	HTTP	417	GET /favicon.ico HTTP/1.1
191	7.283879276	34.223.124.45	10.10.69.28	HTTP	178	HTTP/1.1 200 OK (PNG)
924	51.932058191	10.10.69.28	34.107.221.82	HTTP	355	[TCP Previous segment not captured] GET /canonical.html HTTP/1.1
951	52.087702268	34.107.221.82	10.10.69.28	HTTP	144	HTTP/1.1 200 OK (text/html)
954	52.089322653	10.10.69.28	34.107.221.82	HTTP	357	[TCP Previous segment not captured] GET /success.txt?ipv4 HTTP/1.1
960	52.167599627	34.107.221.82	10.10.69.28	HTTP	62	HTTP/1.1 200 OK (text/plain)
1543	91.971759202	10.10.69.28	185.125.190.48	HTTP	141	GET / HTTP/1.1

```
▸ Frame 124: 398 bytes on wire (3184 bits), 398 bytes captured (3184 bits) on interface eno1, id 0
▸ Ethernet II, Src: 28:c5:c8:bd:c1:01 (28:c5:c8:bd:c1:01), Dst: HewlettP_ba:4a:01 (d8:94:03:ba:4a:01)
▸ Internet Protocol Version 4, Src: 10.10.69.28, Dst: 34.223.124.45
▸ Transmission Control Protocol, Src Port: 58000, Dst Port: 80, Seq: 1, Ack: 1, Len: 344
▾ Hypertext Transfer Protocol
  ▾ GET / HTTP/1.1\r\n
    ▾ [Expert Info (Chat/Sequence): GET / HTTP/1.1\r\n]
        [GET / HTTP/1.1\r\n]
        [Severity level: Chat]
        [Group: Sequence]
      Request Method: GET
      Request URI: /
      Request Version: HTTP/1.1
    Host: neverssl.com\r\n
    User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/116.0\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8\r\n
    Accept-Language: en-US,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    \r\n
    [Full request URI: http://neverssl.com/]
    [HTTP request 1/1]
```

i. http method used – GET
ii. http version used – 1.1
iii. Request is sent to the host – neverssl.com\r\n
iv. Requested Documents:-
    online , faivon.ico , canonical.html , success.txt

v. Optional
   1) Other fields
      - **Length –** It is the number of bytes captured for that particular frame
      - **No. -** It is the packet number over all packets (not including display filter)
      - **Document extensions -** .ico is the website icon on tabs
        .html is most likely a webpage
        .txt?ipv4 is some sort of query
      - **Connection:** Represents the control options for the current network connection. “keep-alive” tell the server to keep the connection open after the response.
      - **Accept-Language –** refers to the client’s preferred languages for the server’s response. In this case U.S English has the highest preference and French has the lowest.

2) Source IP address – 10.10.69.28
   Destination IP address - 34.223.124.45
3) Source Port number – 58000
   Destination Port number – 80

c) Response packets -

http

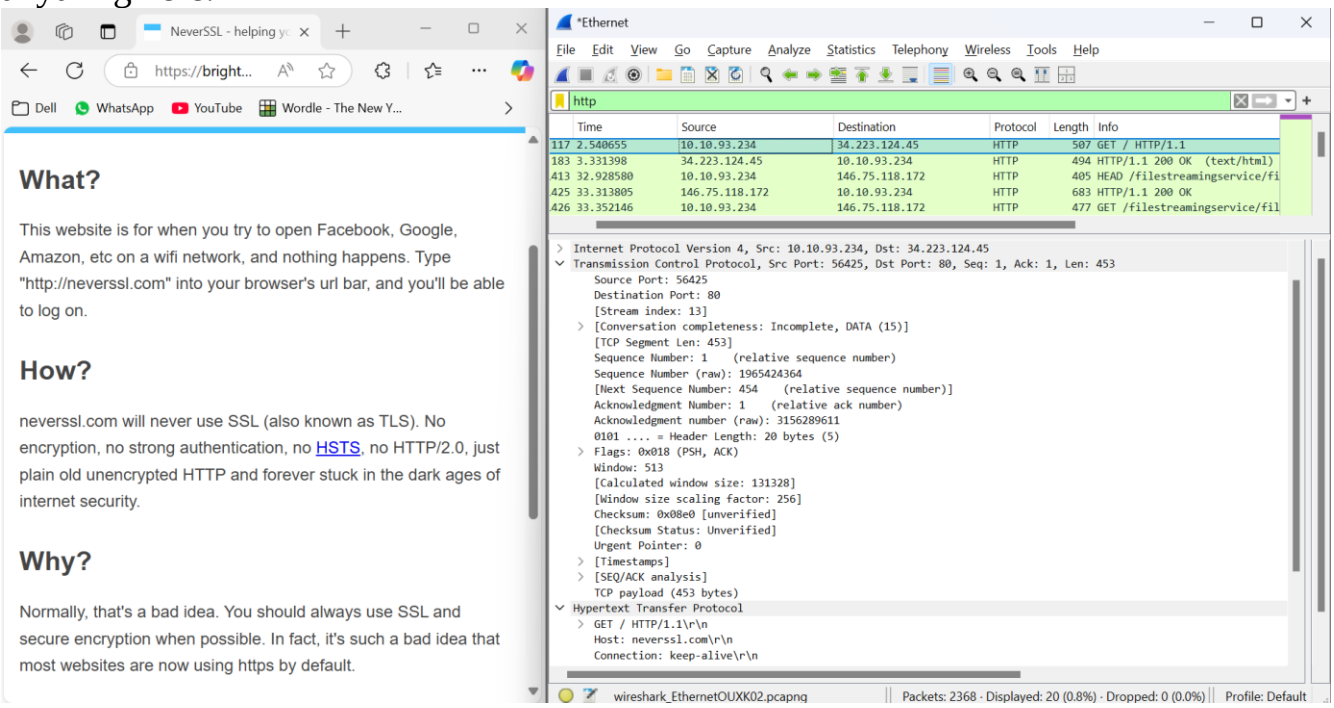
No.	Time	Source	Destination	Protocol	Length	Info
124	5.608595103	10.10.69.28	34.223.124.45	HTTP	398	GET / HTTP/1.1
149	5.711041772	10.10.69.28	34.223.124.45	HTTP	465	GET /online HTTP/1.1
177	6.528463275	34.223.124.45	10.10.69.28	HTTP	319	HTTP/1.1 301 Moved Permanently (text/html)
179	6.531418794	10.10.69.28	34.223.124.45	HTTP	466	GET /online/ HTTP/1.1
183	6.889958364	34.223.124.45	10.10.69.28	HTTP	1227	HTTP/1.1 200 OK (text/html)
185	6.926564887	10.10.69.28	34.223.124.45	HTTP	417	GET /favicon.ico HTTP/1.1
191	7.283879276	34.223.124.45	10.10.69.28	HTTP	178	HTTP/1.1 200 OK (PNG)
924	51.932058191	10.10.69.28	34.107.221.82	HTTP	355	[TCP Previous segment not captured] GET /canonical.html HTTP/1.1
951	52.087702268	34.107.221.82	10.10.69.28	HTTP	144	HTTP/1.1 200 OK (text/html)
954	52.089322653	10.10.69.28	34.107.221.82	HTTP	357	[TCP Previous segment not captured] GET /success.txt?ipv4 HTTP/1.1
960	52.167599627	34.107.221.82	10.10.69.28	HTTP	62	HTTP/1.1 200 OK (text/plain)
1543	91.971759202	10.10.69.28	185.125.190.48	HTTP	141	GET / HTTP/1.1

```
▸ Transmission Control Protocol, Src Port: 80, Dst Port: 58012, Seq: 897, Ack: 824, Len: 1173
▸ [2 Reassembled TCP Segments (1518 bytes): #182(345), #183(1173)]
▾ Hypertext Transfer Protocol
  ▾ HTTP/1.1 200 OK\r\n
    ▾ [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
        [HTTP/1.1 200 OK\r\n]
        [Severity level: Chat]
        [Group: Sequence]
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
    Date: Tue, 21 Jan 2025 10:52:03 GMT\r\n
    Server: Apache/2.4.62 ()\r\n
    Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT\r\n
    ETag: "8be-5e28b29291e10-gzip"\r\n
    Accept-Ranges: bytes\r\n
    Vary: Accept-Encoding\r\n
    Content-Encoding: gzip\r\n
  ▸ Content-Length: 1173\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    X-Cache: MISS from localhost\r\n
    Connection: keep-alive\r\n
```

i. http method – Not found / None
ii. http version – HTTP/1.1
iii. Status code in the response – 200 OK
   It means that the request was successful, and the server returned the requested resource.
   In many cases it doesn't respond with anything, while in this case it responds with GET / HTTP/1.1
iv. Optional:
   1) Other fields in the http response –
      - **Severity level –** It refers to the level of importance of an event. In this case it was "Chat" that means informal or non-critical.
      - **Group –** It is a grouping mechanism to organize certain events. Sequence here refers to 'data in transmission'
      - **Etag –** This is used by web servers to give a specific identifier to a resource being requested. It also represents the contents of the resource
      - **Accept-Ranges –** It represents whether a server allows a client to download a portion of its resource also referred to as "Range-Request". Here it is bytes\r\n meaning that it supports.

2) Source IP address – 34.223.124.45
   Destination IP address – 10.10.69.28
3) Source Port number – 80
   Destination Port number – 58012

d) We observe that in reaction to the response from the server the client doesn't do anything here.



i. http method used – GET
ii. http version used – 1.1
iii. Request is sent to host – neverssl.com\r\n
iv. Requested documents – filestreamingservice/files/ ….
v. Optional –
   1) Same as as b) v. (1)
   2) Source IP address – 10.10.93.234
      Destination IP address – 34.223.124.45
   3) Source Port number – 56425
      Destination Port number – 80

e) In this case the server responds with an status – OK
i. http method used – GET
ii. http version – 1.1
iii. Status code – OK
   Usually the client/source doesn't respond to this
iv. Optional:
   1) Same as c) iv. (1)
   2) Source IP address – 34.223.124.45

Destination IP address – 10.10.93.234

3) Source Port number – 80
   Destination Port number – 56425

f) In the homework section on separate request was sent to /favicon.ico, however in the Lab request to /favicon.ico was sent. This is due to different browsers used. The reason is that **favicon.ico** represents the small icon of the tab section or the bookmark section. This is unique to every website and we also notice that the server responds with an OK (PNG). Represent successful response and the file type being a PNG meaning that it's an image.